

1. Streaming Input

incoming sample (x_t, y_t, a_t)

stream order t

predict $\hat{y}_t$

observe y_t

test-then-train (prequential)

2. Fair Oblique Forest

M soft-routed oblique trees

$$f(x) = \frac{1}{M} \sum_m n_m(x)$$

prediction

- Accuracy
- DP
- Loss
- EO

node stats

output $\hat{y}_t = \arg \max f(x)$

3. Performance Monitor

window W

errors

4. ADWIN Detector

drift warn

change pt

Hoeffding bound on rolling error

5. Reaction Controller

Warning

- pre-warm LR
- arm detectors

Confirmed Drift

- spike LR $\beta_c \eta_0$
- sharpen $\tau \rightarrow \tau_{\text{drift}}$
- reset detectors

Recovery

- anneal LR $\rightarrow \eta_0$
- anneal $\tau \rightarrow 1$

6. Training Update

sample (x_t, y_t, a_t)

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{fair}}$$

- node fairness penalty
- gradient step (Adam) at η_t

7. Fairness Monitor

window W'

- Demographic Parity
- Equalized Odds
- Equal Opportunity

Legend

- Model / data flow (predictions, learning, metrics)
- Drift signals & control flow (detection, reaction)